

Polynomial Regression

Study Plan & Resources

Everything you need to understand polynomial regression and master regression metrics — MSE, MAE, MAPE, RMSE, and R^2 .

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Study Topics

Part 1

Core Polynomial Regression Topics

The fundamental concepts every student must understand before fitting curves and evaluating models...

Topic	What to Know
Linear vs. Polynomial Regression	Understanding why linear regression fails for non-linear data — and how polynomial regression extends it by adding higher-degree feature terms (x^2 , x^3 , ...).
Feature Engineering	Using <code>PolynomialFeatures</code> from <code>sklearn.preprocessing</code> to transform input features into a higher-dimensional space, then feeding them into a <code>LinearRegression</code> model.
Degree Selection	Choosing the right polynomial degree — too low underfits, too high overfits. Understanding the bias-variance tradeoff as degree increases.
Overfitting & Underfitting	Diagnosing underfitting (high training error) and overfitting (low training error, high validation error) using learning curves and validation curves.
Regularization	Ridge (L2) and Lasso (L1) regression as solutions to overfitting in polynomial models — penalizing large coefficients to reduce model complexity.
Train / Validation Split	Using <code>train_test_split</code> and <code>cross_val_score</code> to evaluate model generalization — never fit on the test set.
Pipeline Pattern	Building a clean <code>sklearn</code> Pipeline that chains <code>PolynomialFeatures</code> → <code>StandardScaler</code> → <code>LinearRegression</code> for reproducible, leak-free preprocessing.

Part 2

Regression Metrics (MSE, MAE, MAPE, RMSE, R²)

How to measure how good (or bad) your model is — and when to use each metric...

Topic	What to Know
MSE — Mean Squared Error	Average of squared differences between predicted and actual values. Heavily penalizes large errors. Use when large deviations are especially costly. Formula: $\left(\frac{1}{n}\right) \sum (\hat{y} - y)^2$
MAE — Mean Absolute Error	Average of absolute differences between predicted and actual values. More robust to outliers than MSE, and easier to interpret (same units as y). Formula: $\left(\frac{1}{n}\right) \sum \hat{y} - y $
RMSE — Root Mean Squared Error	Square root of MSE — puts the error back in the same units as y for interpretability, while still penalizing large errors more. Formula: $\sqrt{MSE}$
MAPE — Mean Absolute % Error	Average of absolute percentage errors. Useful for comparing models across different scales. Breaks down when actual values are near zero. Formula: $\left(\frac{1}{n}\right) \sum (\hat{y} - y)/y \times 100\%$
R² — Coefficient of Determination	Measures what fraction of the variance in y is explained by the model. $R^2 = 1$ means perfect fit; $R^2 = 0$ means the model is no better than predicting the mean. Can go negative for very poor models.

Adjusted R ²	R² adjusted for the number of predictors — penalizes adding unnecessary features. Essential for comparing polynomial models of different degrees fairly. NOTE: look it up online, here is a link to help you (this has both R² and Adjusted R²)
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Part 3

Important Use Cases

How polynomial regression and its metrics connect to real modelling, reporting, and analysis workflows...

Once you know the tools, you need to know when and why to use them. These are the key workflows that matter in practice:

	Model Evaluation & Selection	Calculating all five metrics side-by-side for models of different degrees — choosing the degree where validation MSE/RMSE bottoms out and R ² plateaus.
	Overfitting Diagnosis	Plotting training vs. validation RMSE or R ² as degree increases to visually identify the point where the model starts memorizing instead of generalizing.
	Scale-Aware Comparisons	Using MAPE when comparing models trained on datasets with very different y scales, and falling back to MAE or RMSE when actuals contain zeros.
	Reporting Results	Reporting RMSE alongside R ² in a final model summary — RMSE gives interpretable error magnitude, R ² gives relative model quality in one number.
	Pipeline Experiments	Running cross-validated GridSearchCV over polynomial degree and regularization strength (alpha for Ridge) using neg_mean_squared_error as the scoring metric.
	Residual Analysis	Plotting residuals (ŷ – y) vs. fitted values to check for heteroscedasticity — a pattern in residuals means the model is missing structure.

Resources

Part 4

Resources

Curated materials to help you learn and go deeper — start with the team's own content first...

From the AI Sub-Team

NOTEBOOK Regression Metrics Deep Dive Notebook

Our team's Python notebook — implements all five metrics from scratch and compares them side by side on real datasets. [\[Colab Link\]](#)

Playlists & External Guides

Use the scikit-learn documentation — it's the single best reference. docs.scikit-learn.org

VIDEO Polynomial Regression

Fast-paced tutorial covering polynomial regression, overfitting, and regularization with numpy and sklearn. Great for a quick review.

Video by NeuralNine

[\[Polynomial Regression\]](#)

[\[Polynomial regression – Arabic \(by Ahmead Alseirafi\)\]](#)

VIDEO Evaluation Metrics For Regression

Comprehensive video comparing MSE, MAE, RMSE, MAPE, and R^2 with intuitive explanations, formulas, and guidance on when to use each.

Video by NeuralNine

[\[Regression Metrics Explained\]](#)